Radiometry-Aware Evaluation of Cross-Sensor Satellite Image Super-Resolution

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***Abstract*—Real cross-sensor satellite image super-resolution (SR) differs from synthetic SR because the source and target observations vary not only in spatial resolution but also in sensor-dependent radiometry. Consequently, improvements over conventional bicubic interpolation may reflect radiometric adaptation rather than genuine spatial-detail recovery. This paper investigates this distinction using FLAIR-HUB pairs from Sentinel-2, SPOT, and 0.20 m aerial imagery at resolution ratios of $6.4\times$, $8\times$, and $51.2\times$. Representative convolutional, residual, attention, transformer, and diffusion-based SR models are evaluated against both raw and symmetrically radiometrically matched bicubic controls. Results show that apparent PSNR gains of up to 15.01 dB against raw bicubic interpolation become close to zero after radiometric matching, demonstrating that a substantial fraction of the conventional gain originates from cross-sensor radiometric differences. Nevertheless, the SR models recover measurable spatial information, reaching up to $5.55\times$ the sharpness of bicubic interpolation and recovering approximately 13–22% of target gradient magnitude across structured land-cover classes. Incorporating target-resolution normalized digital surface model (nDSM) information in a focused SPOT-to-aerial case study moves the mean EDSR matched-PSNR difference from -0.18 dB to $+0.02$ dB, i.e., approximate parity with the matched control, while increasing sharpness from $3.08\times$ to $3.75\times$, improving LPIPS, and improving detail recovery in seven of eight evaluated classes with approximately 1% additional parameters. These results indicate that real cross-sensor SR should be evaluated jointly through radiometrically matched fidelity, spatial-detail recovery, and perceptual quality rather than raw PSNR alone.**

***Index Terms*—Cross-sensor super-resolution, remote sensing, satellite imagery, radiometric matching, spatial-detail recovery, elevation guidance, FLAIR-HUB.**

I. Introduction

High-resolution Earth observation (EO) imagery is important for land-cover mapping, urban monitoring, agriculture, and environmental assessment. However, satellite sensors involve inherent trade-offs among spatial resolution, spectral resolution, revisit frequency, and coverage. Image super-resolution (SR) provides a computational approach for enhancing spatial information from lower-resolution observations. Deep SR has progressed from convolutional and residual architectures such as SRCNN and EDSR [1], [2] to attention- and transformer-based models such as RCAN and SwinIR [3], [4], while recent remote-sensing studies increasingly explore diffusion and prior-guided reconstruction [5], [6].

A major challenge arises when SR is applied to images acquired by different sensors. Unlike synthetic SR, where the low-resolution image is generated directly from the high-resolution target, real cross-sensor observations can differ in spatial sampling, spectral response, radiometry, illumination, acquisition time, and geometric alignment. Recent datasets and cross-platform studies have therefore shifted attention toward real satellite–aerial SR [7], [8]. Under such conditions, an improvement in conventional fidelity metrics may reflect adaptation to the target sensor's radiometry rather than genuine recovery of missing spatial information.

Building on recent evaluation and radiometric-consistency work [9], [10], we investigate a complementary question: *how much of the apparent improvement of cross-sensor SR originates from radiometric adaptation, and how much corresponds to measurable spatial-detail recovery?*

To answer it, we compare SR predictions against both raw bicubic interpolation and a symmetrically radiometrically matched bicubic control. Pixel fidelity is further complemented by perceptual similarity and gradient-based spatial-detail measurements. Experiments are conducted on FLAIR-HUB [11] using three real cross-sensor configurations: Sentinel-2→SPOT ($6.4\times$), SPOT→aerial ($8\times$), and Sentinel-2→aerial ($51.2\times$). Representative convolutional, residual, attention, transformer, and diffusion-based models are evaluated under geographically disjoint training and test regions.

The main contributions of this work are threefold. First, we conduct a real cross-sensor SR study spanning Sentinel-2, SPOT, and aerial imagery, with resolution gaps from $6.4\times$ to $51.2\times$ under geographically disjoint evaluation. Second,

TABLE I
POSITIONING AGAINST RECENT REAL-WORLD REMOTE-SENSING SR WORK

Work	Real cross-sensor	Matched radiom. control	Spatial-detail metric	Land-cover analysis	nDSM
SEN2NAIP [7]	✓	–	–	–	–
OpenSR-test [9]	✓	–	✓	–	–
SEN2SR [10]	✓	–	–	–	–
Cross-platform [8]	✓	–	–	–	–
This paper	✓	✓	✓	✓	✓

we introduce a radiometrically controlled evaluation that separates apparent SR gains from radiometric adaptation and jointly considers matched fidelity, perceptual similarity, and spatial-detail recovery, including land-cover-specific analysis. Third, in a focused SPOT→aerial case study, we investigate an elevation-guided SR refinement using target-resolution normalized digital surface model (nDSM) data, which improves detail recovery in seven of eight evaluated land-cover classes while adding approximately 1% to the EDSR parameter count. The contribution is primarily an evaluation methodology; the SR backbones are established architectures, and no new general-purpose SR network is proposed.

II. Related Work

Real-world remote-sensing SR has moved from synthetically degraded pairs toward real cross-sensor observations. SEN2NAIP pairs Sentinel-2 with aerial NAIP imagery [7], and cross-platform diffusion SR enhances satellite imagery with aerial targets [8]. OpenSR-test evaluates SR through consistency, synthesis, and correctness rather than full-reference fidelity alone [9], and SEN2SR enforces radiometric and spatial consistency with the Sentinel-2 input [10]. Table I summarizes how the present study complements these works: rather than proposing a new benchmark or SR network, it measures how much of a reported cross-sensor gain survives an explicit, symmetrically applied radiometric control, and relates the remainder to spatial-detail, land-cover, and elevation-guided analyses.

III. Proposed Radiometry-Aware Cross-Sensor SR Framework

The framework is summarized in Fig. 1.

Let $\mathbf{x} \in \mathbb{R}^{H\times W\times C}$ denote a low-resolution source-sensor observation and $\mathbf{y} \in \mathbb{R}^{sH\times sW\times C}$ the corresponding high-resolution target-sensor image, where s is the cross-sensor resolution ratio. The source image is first interpolated to the target grid,

$$\mathbf{x}_{\mathrm{bic}} = \mathcal{U}_s(\mathbf{x}), \tag{1}$$

where $\mathcal{U}_s(\cdot)$ denotes bicubic interpolation. The regression models predict a correction to this interpolated input,

$$\hat{\mathbf{y}} = \mathbf{x}_{\mathrm{bic}} + f_\theta(\mathbf{x}), \tag{2}$$

where $f_\theta(\cdot)$ is the trainable SR model and $\hat{\mathbf{y}}$ is the reconstructed target-resolution image. SRCNN, RCAN, and SwinIR apply f_θ to $\mathbf{x}_{\mathrm{bic}}$, whereas EDSR applies it to $\mathbf{x}$ and upsamples internally; the conditional DDPM instead samples $\hat{\mathbf{y}}$ conditioned on the source observation.

A direct comparison between $\hat{\mathbf{y}}$ and $\mathbf{y}$ mixes spatial reconstruction with differences in sensor radiometry. Therefore, in addition to the raw bicubic control, a radiometrically matched control is computed. For any image $\mathbf{z} \in \{\mathbf{x}_{\mathrm{bic}}, \hat{\mathbf{y}}\}$ and channel c, a least-squares gain and offset onto the target are estimated independently for every test image,

$$(\alpha_c(\mathbf{z}), \beta_c(\mathbf{z})) = \arg\min_{\alpha,\beta} \|\alpha \mathbf{z}_c + \beta - \mathbf{y}_c\|_2^2, \tag{3}$$

and the matched image is $\mathcal{M}(\mathbf{z})_c = \mathrm{clip}_{[0,1]}(\alpha_c(\mathbf{z})\mathbf{z}_c + \beta_c(\mathbf{z}))$. The bicubic control and the SR prediction each receive their own optimal fit, $\tilde{\mathbf{x}}_{\mathrm{bic}} = \mathcal{M}(\mathbf{x}_{\mathrm{bic}})$ and $\tilde{\mathbf{y}} = \mathcal{M}(\hat{\mathbf{y}})$. The control is therefore symmetric: both outputs are given the same oracle radiometric correction, so the SR model receives no fitting advantage that the bicubic control lacks. Ignoring clipping, the matched squared error of channel c equals $\sigma_{y_c}^2(1-\rho_c^2)$, where ρ_c is the Pearson correlation between $\mathbf{z}_c$ and $\mathbf{y}_c$; matched PSNR is thus invariant to any per-channel gain and offset of either output and compares only how well its spatial structure correlates with the target. Because the fit uses $\mathbf{y}$, the matched control is an *oracle evaluation control*: it is computed only at evaluation time and is never available to, or used by, any model during training or inference. It is not an operational preprocessing step. The same procedure and code are applied identically to every model and all three sensor configurations.

The affine model is chosen deliberately. A per-channel gain and offset is the first-order description of inter-sensor differences in calibration, atmospheric conditions, and illumination; it has a closed-form solution; and, being a global per-channel mapping, it cannot add spatial structure. Hence, an advantage remaining after matching cannot be explained by a global linear radiometric difference. The model also has limitations. Spectral-response mismatch, nonlinear radiometric relations, spatially varying illumination and shadows, and acquisition-time differences are not captured by a per-image affine map, so part of the cross-sensor discrepancy can remain after matching. $\Delta_{\mathrm{matched}}$ should therefore be read as a first-order radiometric control rather than a complete removal of sensor effects. Two complementary PSNR differences are reported:

$$\Delta_{\mathrm{raw}} = \mathrm{PSNR}(\hat{\mathbf{y}}, \mathbf{y}) - \mathrm{PSNR}(\mathbf{x}_{\mathrm{bic}}, \mathbf{y}), \tag{4}$$

and

$$\Delta_{\mathrm{matched}} = \mathrm{PSNR}(\tilde{\mathbf{y}}, \mathbf{y}) - \mathrm{PSNR}(\tilde{\mathbf{x}}_{\mathrm{bic}}, \mathbf{y}). \tag{5}$$

Here, Δ_{raw} measures the apparent improvement over the conventional interpolation baseline, whereas $\Delta_{\mathrm{matched}}$ measures the remaining fidelity difference after symmetric radiometric correction. The gap between them is used as an empirical diagnostic of how strongly the apparent gain depends on radiometric differences that a global affine map can absorb. It is not an exact decomposition of the SR gain into "radiometric" and "spatial" components, because PSNR is not additive and the two effects interact.

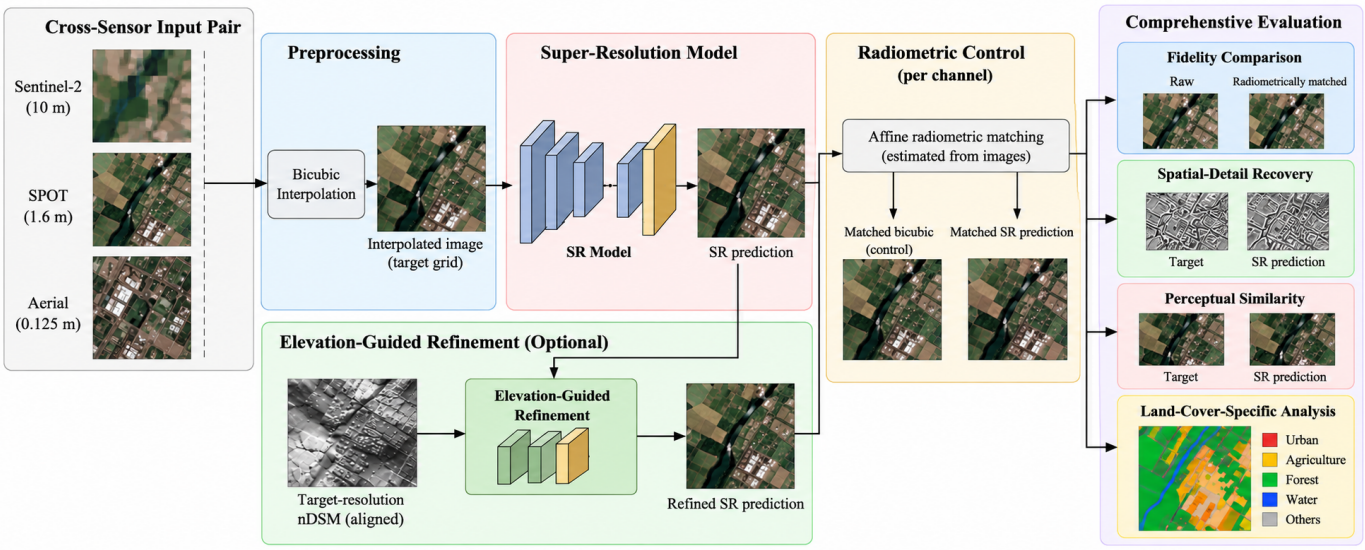


Fig. 1. Overview of the proposed cross-sensor SR framework. The low-resolution observation is reconstructed by an SR model and evaluated using raw and radiometrically matched controls to separate apparent radiometric gains from spatial-detail recovery. Perceptual and gradient-based measures provide complementary evaluation, while target-resolution nDSM information can be incorporated through a lightweight elevation-guided refinement module.

Because matched PSNR alone does not quantify the amount of recovered high-frequency information, spatial detail is evaluated using image gradients. For an image $\mathbf{z}$ containing N pixels, the mean gradient magnitude is defined as

$$G(\mathbf{z}) = \frac{1}{2}\left(\overline{|\partial_x \mathbf{z}|} + \overline{|\partial_y \mathbf{z}|}\right), \tag{6}$$

where ∂_x, ∂_y are forward finite differences and the bar denotes the mean over all pixels and channels. The target-normalized spatial-detail recovery is

$$R_{\text{detail}}(\mathbf{z}) = \frac{G(\mathbf{z})}{G(\mathbf{y})}, \tag{7}$$

and *sharpness* relative to interpolation is

$$S = \frac{\overline{R}_{\text{detail}}(\hat{\mathbf{y}})}{\overline{R}_{\text{detail}}(\mathbf{x}_{\text{bic}})} \approx \frac{G(\hat{\mathbf{y}})}{G(\mathbf{x}_{\text{bic}})}, \tag{8}$$

where $\overline{R}_{\text{detail}}$ is averaged over the test set. The first quantity indicates the fraction of target gradient magnitude represented by the reconstruction, whereas S quantifies the increase in high-frequency content relative to bicubic interpolation ($S = 1$ for bicubic itself). Throughout the paper, *gradient magnitude* refers to the absolute quantity $G(\cdot)$, and *sharpness* refers to the dimensionless ratio S; the two are not used interchangeably. Sharpness is a proxy for recovered spatial detail because bicubic interpolation cannot create frequencies above the source Nyquist limit, so additional gradient energy must come from the model. It measures the *amount* of high-frequency content, not its *correctness*: ringing, noise, and hallucinated texture also increase G. It is therefore interpreted only together with matched fidelity, LPIPS, and the qualitative examples. Gradient magnitude is insensitive to additive intensity offsets but scales with multiplicative radiometric gain; therefore, it is interpreted jointly with the radiometrically matched fidelity analysis rather than as a radiometrically invariant metric.

To determine whether recovery depends on scene content, the same measure is computed over pixels belonging to land-cover class k,

$$R_k = \frac{G_k(\hat{\mathbf{y}})}{G_k(\mathbf{y})}, \tag{9}$$

where $G_k(\cdot)$ denotes the mean gradient magnitude restricted to class k, computed as the ℓ_2 norm of central differences of the channel-mean image and pooled over all class-k pixels of the evaluated tiles. Reporting $G_k(\mathbf{y})$ together with R_k is important because a high recovery percentage may correspond to only a small amount of absolute detail when the target class itself contains little high-frequency structure.

Finally, target-resolution elevation information is incorporated as an auxiliary geometric prior. Let $\mathbf{h}$ denote the normalized nDSM aligned with the target image. A lightweight refinement network $g_\phi(\cdot)$ predicts a residual correction from the concatenated SR output and elevation map,

$$\hat{\mathbf{y}}_{\text{EG}} = \hat{\mathbf{y}} + g_\phi\left([\hat{\mathbf{y}}, \mathbf{h}]\right), \tag{10}$$

where $[\cdot,\cdot]$ denotes channel-wise concatenation. The final convolution of g_ϕ is initialized to zero so that, at initialization, the guided model reduces to its unguided backbone; backbone and refinement are then trained jointly. The refinement module contains only 11,299 trainable parameters, corresponding to approximately 1% of the EDSR parameter count.

Data. All pairs come from FLAIR-HUB [11]. Each tile covers 102.4×102.4 m and is co-registered across Sentinel-2 (10×10 px, 10.24 m), SPOT (64×64 px, 1.6 m), and aerial

imagery (512×512 px, 0.20 m); the aerial target is used at its native resolution without downsampling. All experiments use the three RGB channels: Sentinel-2 B4/B3/B2 as the median of 12 cloud-filtered temporal bins, the first three SPOT bands, and the aerial RGB image. Satellite reflectance is scaled by $1/0.3$ and aerial digital numbers by $1/255$, both clipped to $[0, 1]$. Training uses 4,662 tiles from seven domains (D004, D006, D011, D017, D022, D033, D037) and testing uses 1,332 tiles from two geographically disjoint domains (D040, D044). Every model is trained on the same tiles, and every (architecture, configuration, seed) is trained separately from scratch; no model is transferred between sensor pairs. No separate validation split is used, and the final epoch is evaluated without checkpoint selection.

Training. SRCNN [1] (188k parameters), EDSR [2] (927k), RCAN [3] (1.01M), and SwinIR [4] (241k) are trained with an ℓ_1 loss; the conditional DDPM (264k) uses the noise-prediction MSE loss with $T{=}200$ linear-schedule steps and 200-step ancestral sampling. All models use AdamW with a constant learning rate of 2×10^{-4}, effective batch size 32 (gradient accumulation for 512-px targets), mixed precision, horizontal-flip augmentation, and 25 epochs. Hyperparameters are deliberately identical across architectures rather than individually tuned. RCAN and DDPM are evaluated on Sentinel-2→SPOT only. The nDSM refinement (Eq. 10) is trained jointly with its backbone from scratch using the same settings, with the nDSM converted to meters, divided by 30, and clipped to $[0, 1]$.

Evaluation and uncertainty. PSNR is computed on $[0, 1]$ data over all channels per tile and averaged over the 1,332 test tiles. LPIPS [12] uses the AlexNet variant on RGB inputs scaled to $[-1, 1]$; lower is better, and the *LPIPS improvement* plotted in Fig. 3 is the absolute difference $\mathrm{LPIPS}(\mathbf{x}_{\mathrm{bic}}, \mathbf{y}) - \mathrm{LPIPS}(\hat{\mathbf{y}}, \mathbf{y})$, so positive values favor the SR model. Where five seeds (7, 41, 913, 1337, 2026) are available, values are seed means and $\pm$ denotes the sample standard deviation across seeds of each run's test-set mean; it is neither a per-image spread nor a confidence interval. Cells marked † in Table II are single runs (seed 1337) and carry no uncertainty estimate. Experiments were run on cloud GPUs via Modal.

IV. Results and Discussion

The experimental results reveal a clear distinction between apparent fidelity improvement and spatial-detail recovery in real cross-sensor SR. Figure 2 compares the PSNR improvement obtained against the raw bicubic control with that obtained after symmetric radiometric matching, and Table II lists the underlying absolute values. For Sentinel-2→SPOT, SRCNN, EDSR, and RCAN achieve apparent gains of 14.70–15.01 dB (seed s.d. ≤ 0.21 dB); the SwinIR mean is lower (10.39 dB) because one of its five seeds diverged. The corresponding gains are 6.98–7.29 dB for SPOT→aerial and 1.09–1.36 dB for Sentinel-2→aerial. The absolute values explain these gains: matching alone raises bicubic PSNR from 14.25 to 35.24 dB for Sentinel-2→SPOT and from 10.63 to 21.74 dB for SPOT→aerial, which shows how large the cross-sensor radiometric offset is relative to the spatial error.

TABLE II
Absolute Test-Set Metrics. PSNR in dB; S Is Sharpness Relative to Bicubic (Eq. 8); S2 Denotes Sentinel-2

Config.	Model	PSNR raw	PSNR matched	LPIPS↓	S
S2→ SPOT (6.4×)	Bicubic	14.25	35.24	0.562	1.00
	SRCNN	28.94	34.62	0.334	0.59
	EDSR	29.26	34.93	0.318	1.00
	RCAN	29.16	34.77	0.329	0.65
	SwinIR*	24.63	34.58	0.381	0.91
	DDPM†	26.00	33.59	0.288	3.58
SPOT→ aerial (8×)	Bicubic	10.63	21.74	0.913	1.00
	SRCNN	17.68	21.56	0.851	1.90
	EDSR	17.92	21.57	0.868	3.22
	SwinIR†	17.62	21.40	0.839	5.55
S2→ aerial (51.2×)	Bicubic	16.44	21.58	0.871	1.00
	SRCNN†	17.53	20.99	0.849	1.43
	EDSR†	17.80	21.33	0.883	2.43
	SwinIR†	17.56	20.94	0.845	3.58

†Single seed. *Mean includes one diverged seed (raw-PSNR s.d. 10.5 dB).
Other model rows: mean of five seeds; raw-PSNR s.d. ≤ 0.21 dB, matched s.d. ≤ 0.13 dB.

After symmetric radiometric matching, however, the fidelity advantage of the regression-based configurations becomes close to zero and generally slightly negative. This large gap indicates that most of the apparent PSNR gain can be reproduced by a global per-channel affine correction of the bicubic control, i.e., it is largely attributable to cross-sensor radiometric differences rather than spatial reconstruction. Raw PSNR against bicubic interpolation can therefore substantially overstate the spatial benefit of a cross-sensor SR model.

Importantly, the small matched-PSNR differences do not imply that the models fail to reconstruct spatial information. As shown in Fig. 3, the SR outputs contain substantially more high-frequency content than bicubic interpolation, and the amount depends strongly on architecture. For SPOT→aerial, SwinIR reaches $5.55\times$ the bicubic sharpness (single seed), followed by EDSR at $3.22 \pm 0.27\times$ and SRCNN at $1.90 \pm 0.36\times$. For Sentinel-2→aerial, the corresponding single-seed ratios are $3.58\times$, $2.43\times$, and $1.43\times$. In the smaller Sentinel-2→SPOT gap, no regression model exceeds the control: EDSR matches it ($1.00\times$), SwinIR falls slightly below ($0.91\times$), and RCAN and SRCNN produce smoother outputs at $0.65\times$ and $0.59\times$. These ratios are relative; in absolute terms, the bicubic control retains only about 5% of the target gradient magnitude $G(\mathbf{y})$ for SPOT→aerial, so even a $5.55\times$ ratio corresponds to roughly a quarter of the target's gradient magnitude.

The diffusion model further illustrates the distinction between perceptual quality and pixel fidelity. For Sentinel-2→SPOT, the conditional DDPM (single seed, one sample per tile) reaches $3.58\times$ the bicubic sharpness and an LPIPS of 0.2880 compared with 0.5615 for the control, while its matched PSNR difference is -1.649 dB. Thus, stronger high-frequency content and improved perceptual similarity do not necessarily imply improved pixel-wise fidelity, emphasizing the need to evaluate these properties jointly.

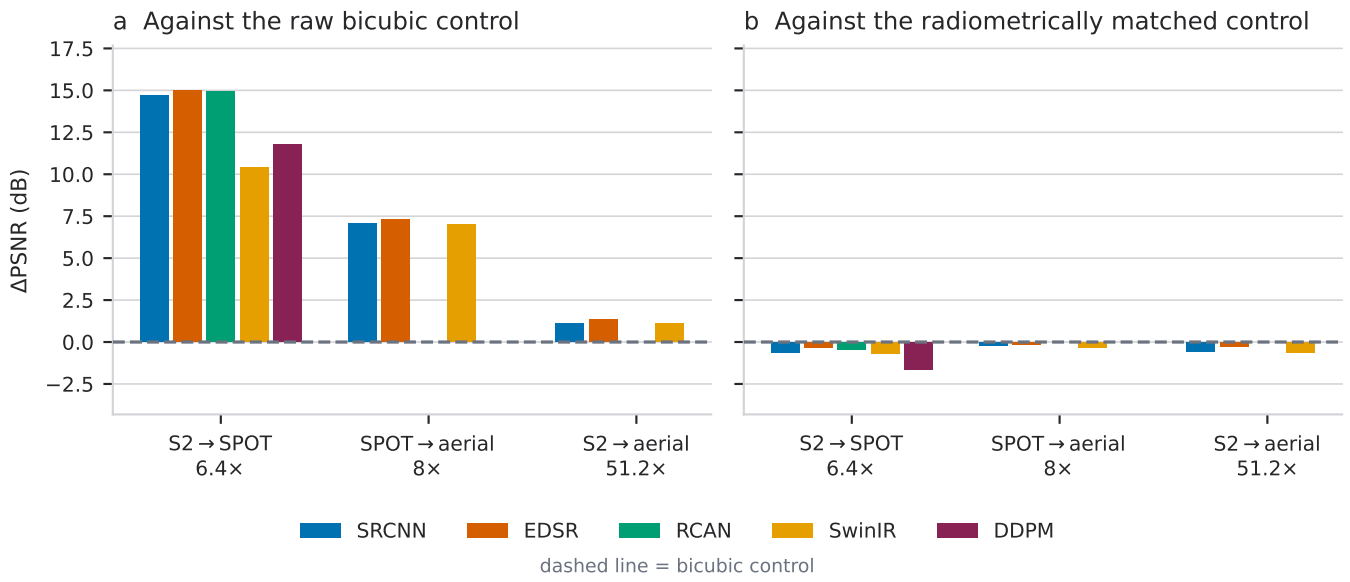


Fig. 2. The same SR models evaluated against two controls. **(a)** Against raw bicubic interpolation, apparent PSNR gains reach approximately 15 dB. **(b)** After symmetric per-image radiometric matching, the advantage becomes close to zero. The contrast between the two evaluations exposes the strong contribution of cross-sensor radiometric differences.

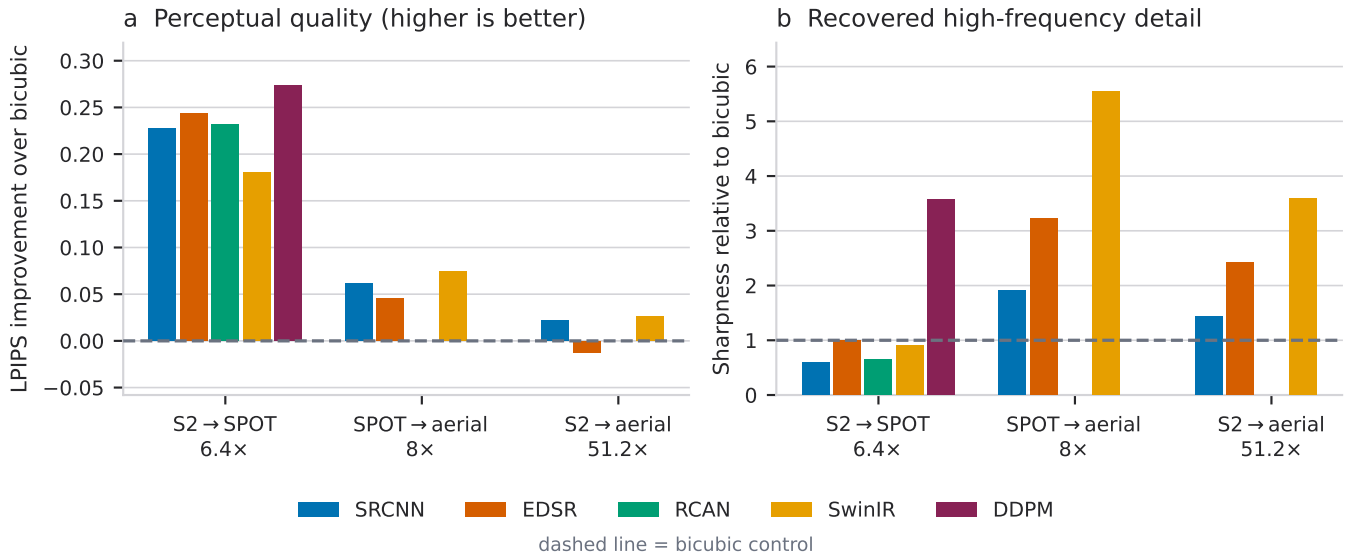


Fig. 3. **(a)** LPIPS improvement relative to the bicubic control, $\mathrm{LPIPS}(\mathbf{x}_{\mathrm{bic}}, \mathbf{y}) - \mathrm{LPIPS}(\hat{\mathbf{y}}, \mathbf{y})$ (higher is better). **(b)** Output sharpness S relative to bicubic interpolation (Eq. 8). The results show that models can recover substantial high-frequency content even when their radiometrically matched PSNR improvement is small or negative.

The land-cover analysis in Fig. 4 further shows that detail recovery depends on scene content. Class masks are the FLAIR-HUB land-cover labels on the 0.20 m aerial grid. The analysis uses SPOT→aerial EDSR and EDSR+nDSM on 300 test tiles, pooling every class pixel across tiles, and reports the eight most frequent classes (each covering at least 0.5% of pixels, i.e., more than 1.7 million pixels per class). Because it uses one trained model per variant, no per-class uncertainty is available, so class-level differences of a few percentage points should be read as indicative rather than statistically established. Across structured vegetated and built classes, EDSR recovers approximately 13–22% of the target gradient magnitude, while the elevation-guided variant reaches approximately 14–24%. Agricultural regions show a particularly large relative improvement: bicubic interpolation recovers only 3.3% of the target gradient magnitude, compared with 15.3% for EDSR.

The recovery fraction must nevertheless be interpreted to-

gether with the target's own gradient magnitude. Water reaches approximately 54% recovery without elevation guidance and 81% with guidance, but its target gradient magnitude is only 0.0070, compared with approximately 0.021–0.032 for more structured classes. Conversely, deciduous vegetation has a target gradient magnitude of 0.0271 but achieves only 13.1% recovery with EDSR. A large normalized recovery fraction therefore does not necessarily correspond to a large amount of absolute spatial information.

The nDSM experiment is a focused case study restricted to the SPOT→aerial configuration and the SRCNN and EDSR backbones; it is not generalized here to the other sensor pairs. Incorporating target-resolution nDSM increases the matched PSNR difference of EDSR from -0.178 ± 0.099 dB to $+0.022 \pm 0.054$ dB, while its sharpness relative to bicubic increases from $3.08\times$ to $3.75\times$. LPIPS simultaneously improves from 0.8696 to 0.7822. SRCNN shows the same behavior: matched PSNR improves from -0.183 ± 0.126 dB to -0.041 ± 0.073 dB, sharpness increases from $1.91\times$ to $2.50\times$, and LPIPS decreases from 0.8521 to 0.7968. The baselines in Table III are the unguided runs paired with the guided ones and therefore differ slightly from the independent sweep in Table II (e.g., EDSR sharpness $3.08\times$ versus $3.22\times$), which is within the seed spread.

The spread of the matched-PSNR difference is comparable to the change in its mean, so the positive mean of EDSR+nDSM should be interpreted as approximate parity with the matched bicubic control rather than statistically established superiority. The shift is nevertheless accompanied by stronger spatial-detail recovery and better perceptual similarity while requiring only 11,299 additional parameters, approximately 1% of the EDSR parameter count. Detail recovery improves in seven of the eight evaluated land-cover classes; for built and impervious regions it increases from approximately 21.5% to 24.0%. The effect is not uniform, however: the plowed class decreases from approximately 15.5% to 13.0%. Within the tested SPOT→aerial setting, nDSM guidance therefore improves several metrics and most classes, but its benefit is class- and scene-dependent.

Fig. 5 provides complementary qualitative evidence. The bicubic reconstructions remain visibly smooth, particularly in regions containing fine texture and narrow spatial structures, whereas the learned SR outputs recover sharper boundaries and additional local detail. The qualitative results are therefore consistent with the quantitative sharpness analysis: the SR models recover measurable spatial structure even when the improvement over a radiometrically matched control is small.

Overall, the experiments demonstrate that no single metric adequately describes real cross-sensor SR. Raw PSNR can be dominated by radiometric adaptation, whereas matched PSNR alone can understate measurable spatial-detail recovery. At the same time, increased sharpness does not guarantee faithful reconstruction. Cross-sensor SR should therefore be evaluated jointly through radiometrically matched fidelity, spatial-detail recovery, perceptual similarity, and scene-dependent analysis.

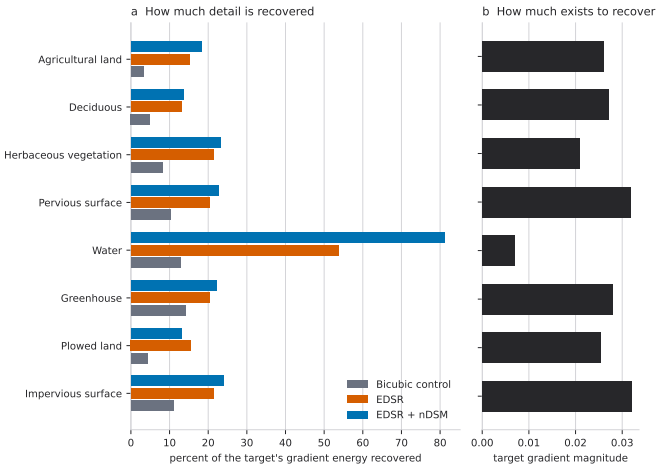


Fig. 4. **(a)** Fraction of target spatial detail recovered by land-cover class. **(b)** Target gradient magnitude for each class. The target-gradient panel is required to interpret normalized recovery values across surfaces with different intrinsic spatial structure.

TABLE III
EFFECT OF TARGET-RESOLUTION NDSM GUIDANCE FOR SPOT→AERIAL SR (MEAN ± S.D. OVER FIVE SEEDS; PAIRED BASELINES TRAINED ALONGSIDE THE GUIDED RUNS)

Method	Δ_{matched} (dB)	Sharpness	LPIPS ↓
SRCNN	-0.183 ± 0.126	$1.91\times$	0.8521
SRCNN+nDSM	-0.041 ± 0.073	$2.50\times$	0.7968
EDSR	-0.178 ± 0.099	$3.08\times$	0.8696
EDSR+nDSM	$\mathbf{+0.022 \pm 0.054}$	$\mathbf{3.75\times}$	**0.7822**

V. CONCLUSION

This paper evaluated real cross-sensor satellite image super-resolution using FLAIR-HUB across resolution gaps of $6.4\times$–$51.2\times$. Results showed that apparent PSNR gains approaching 15 dB over raw bicubic interpolation largely disappear after symmetric radiometric matching. Nevertheless, SR models achieved up to $5.55\times$ higher sharpness and recovered approximately 13–22% of the target gradient magnitude across structured land-cover classes. In a focused SPOT→aerial case study, nDSM guidance brought EDSR to approximate matched-PSNR parity with the control (-0.18 to $+0.02$ dB), increased sharpness from $3.08\times$ to $3.75\times$, and improved most, but not all, land-cover classes with only 1% additional parameters. These findings highlight the importance of jointly evaluating radiometrically matched fidelity, spatial-detail recovery, and perceptual quality rather than relying on raw PSNR alone.

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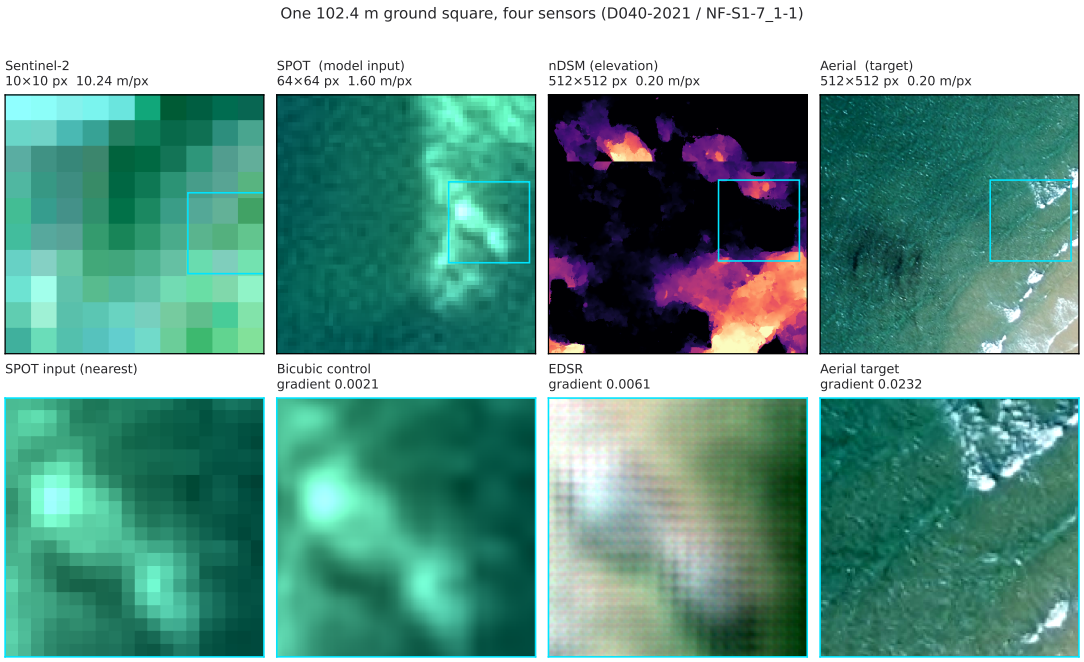


Fig. 5. Representative qualitative cross-sensor SR results on held-out imagery. The comparison illustrates the progression from the low-resolution observation and bicubic control toward the SR reconstruction and high-resolution target.

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